

■ UNSW × MISTRAL AI × ATlassian — 2026 · TEAM KOLLAB

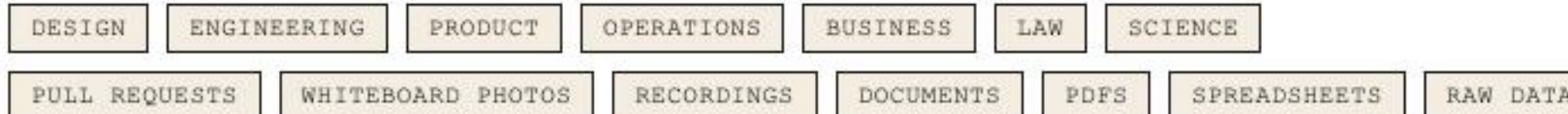
SPATIAL BRAIN

A research paper becomes a Jira issue becomes a pull request — and every hop stays visible on one canvas.

■ THE OFFICIAL PROBLEM STATEMENT

“How might AI help multi-disciplinary teams make sense of information, present ideas, align on decisions, and review work more effectively?”

■ WHAT THE PROBLEM INCLUDES



 WHO THIS IS FOR

PRODUCT MANAGER

Priya Raman

RESEARCH SCIENTIST

Dr Aisha Khan

SOFTWARE ENGINEER

Marco Silva

The pain: Cross-disciplinary work loses its reasoning in the handoff. A scientist's finding is retyped into a ticket, the ticket loses the constraint that made it necessary, and the engineer implements something subtly wrong.

AI should package and pressure-test context — not replace judgment.

■ THE SOLUTION, ONE SENTENCE

A shared semantic canvas where multi-modal ingest and a Mistral specialist roster cover make-sense / present / align / review — while Spatial Brain stays the system of record.

```
Canvas (Next.js) — FastAPI — SQLite (system of record)
|
├─ Multimodal extract → candidates → human promote
├─ Align / Present / Review graph tools
├─ Mistral Agents (Coordinator + specialists)
├─ Lineage assembler → MCP tools → coding agent
└─ Jira outbound + GitHub/Jira inbound webhooks
```

■ FOUR PILLARS — MAPPED TO THE PROBLEM STATEMENT

MAKE SENSE	PDF / whiteboard photo / CSV → candidates → human promote
PRESENT IDEAS	Present Mode + stakeholder brief + generated one-pager
ALIGN ON DECISIONS	Check alignment + decision trail — decided / deferred / rejected
REVIEW WORK	Jira assign + PR badge + constraint checklist + inbound webhooks
+ RECOMMEND	From a finding or constraint, Mistral auto-creates linked task nodes

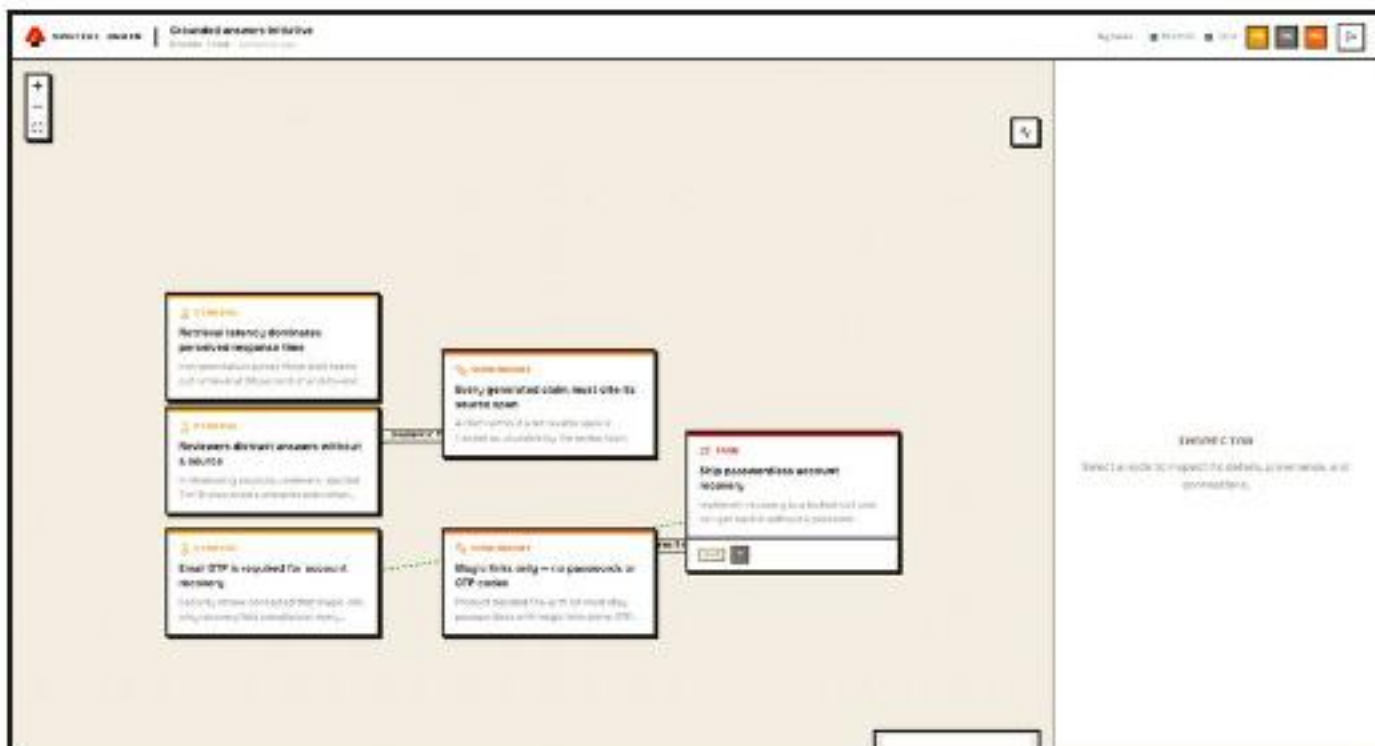
■ LIVE DEMO, SIX BEATS

- 1 Science · PDF — make sense**
Drop the paper. Mistral proposes findings and constraints; promote a few.
- 2 Design · whiteboard — make sense**
Drop a photo of a sketch. Same review UI, different input.
- 3 Ops · spreadsheet — make sense**
Drop a CSV. The data analyst path proposes metric constraints.
- 4 Product · align — align on decisions**
Check alignment on the recovery task — the Arbiter flags the OTP-vs-magic-links contradiction.
- 5 All · present — present ideas**
Present Mode walks claim → quote → implication → task, with a generated one-pager.
- 6 Engineering · review — review work**
Assign to Marco → real Jira issue. Link a PR. Run the constraint checklist.

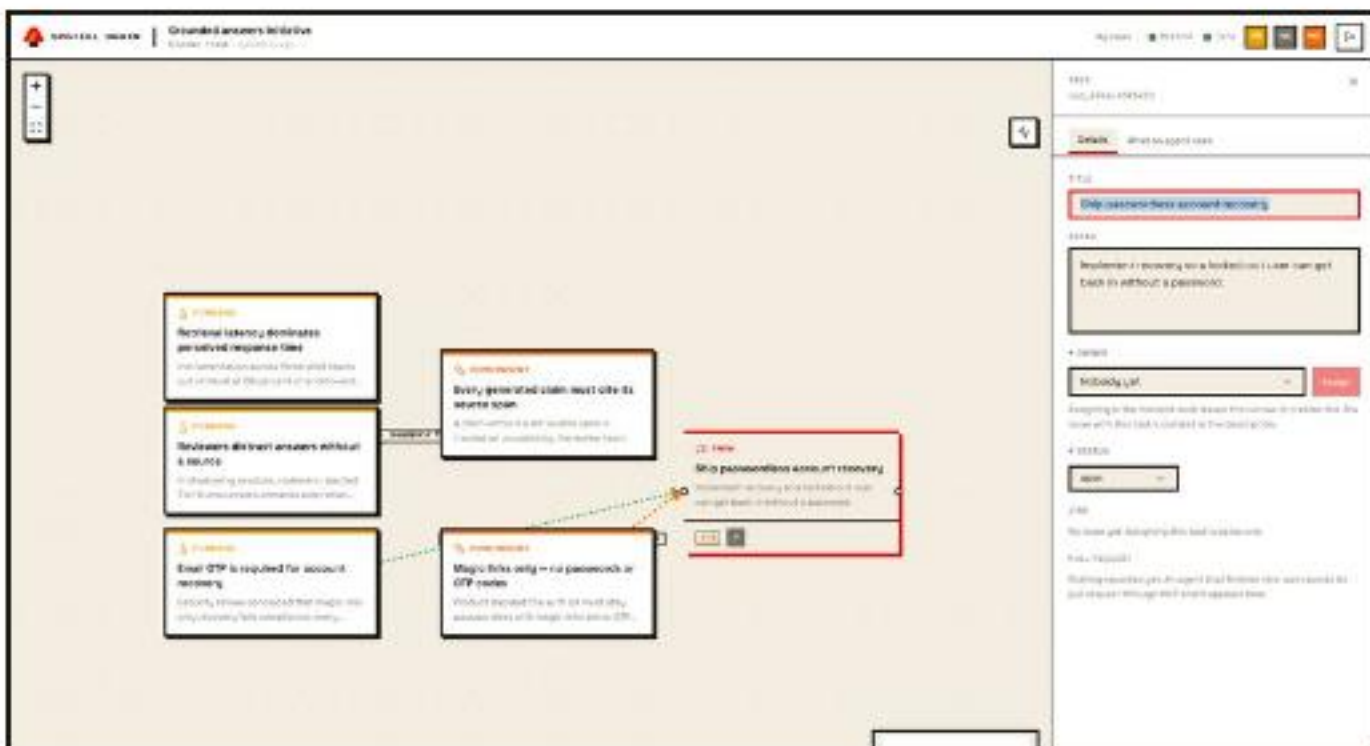
WHAT IT ACTUALLY LOOKS LIKE



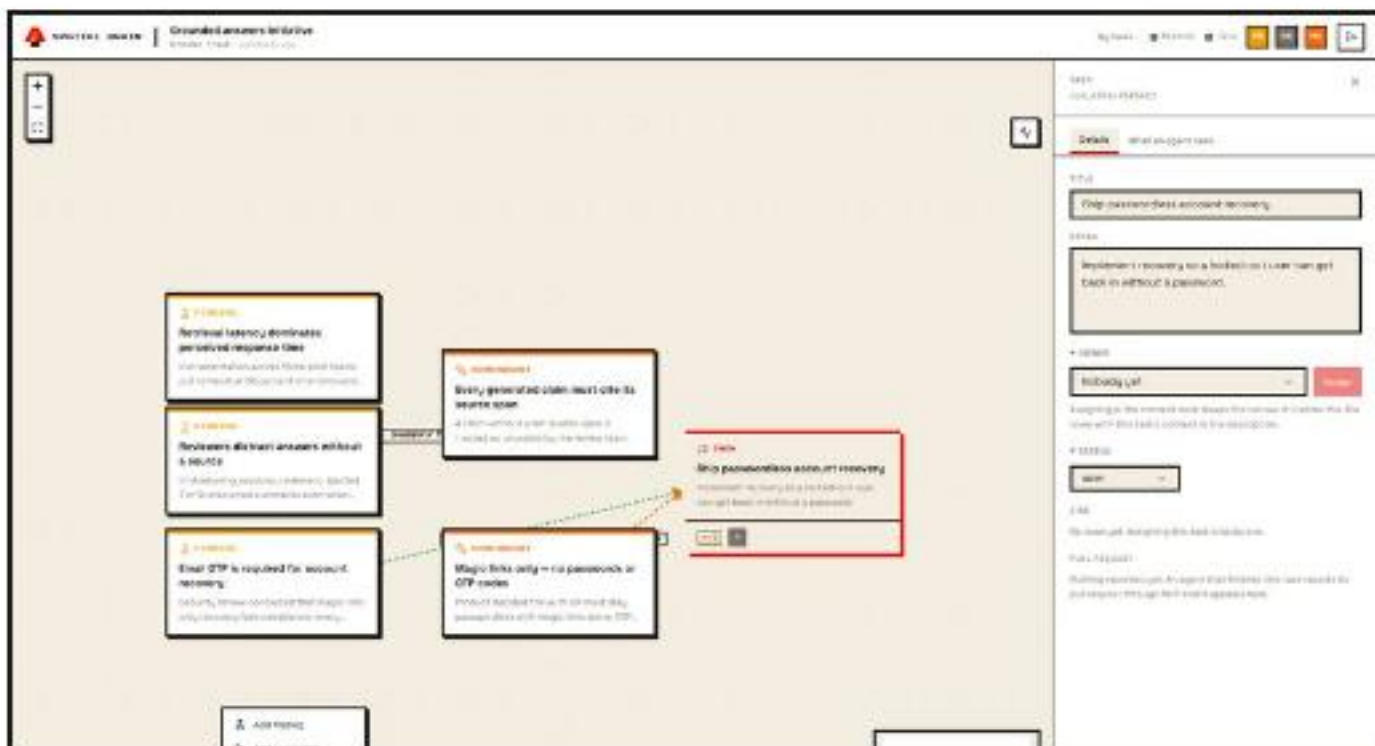
Sign in — one-click persona picker for the demo



The canvas — typed nodes, colour-coded, connected by relation



Inspector — owner, status, Jira sync, PR tracking



Right-click to create — context-aware node creation

 WHY WE'RE DIFFERENT**HEPTABASE**

Human-curated notes with AI assistance layered on top. No agent-facing API — the knowledge stays inside the app.

AGENT-MCP

Built for AI coding agents collaborating in one codebase. Developer-only, not multi-disciplinary humans.

ATLASSIAN'S JIRA MCP

Exposes *existing* Jira data to agents. Doesn't generate the reasoning behind a ticket in the first place.

Spatial Brain: AI extracts automatically, with citations. Multi-disciplinary humans *and* agents. Generates the context, then writes it into Jira — upstream of every competitor listed here.

■ FOUR CLAIMS WE CAN DEFEND LIVE

01 · LINEAGE

A graph walk, not an LLM guessing. Plain BFS in `lineage.py` — Mistral only summarizes a set that's already deterministically selected.

02 · PROVENANCE

Every extracted fact carries a verbatim quote, page number, confidence score, and whether a human or a model asserted it.

03 · THE LOOP CLOSES

An agent doesn't just read context via MCP — it writes back. A reported PR shows up as a badge on the canvas and a Jira comment.

04 · REAL INTEGRATIONS

A real Jira Cloud issue via actual REST calls, with ambiguous-timeout handling — not a mock, not a screenshot.

■ SCORED AGAINST THE OFFICIAL RUBRIC

30%

Value + Human Insight

30%

Technical Execution

20%

Creativity + Design

20%

Feasibility + Scalability

HIDDEN SIGNALS AWARD — “Uncover a non-obvious pattern, contradiction, anomaly, or connection.” That’s *exactly* what Check Alignment does, live, on stage.

■ CLOSING

**Nothing here is a new place to work.
It is the missing edge between the places people already work.**

Spatial Brain — Team Kollab.
UNSW × Mistral AI × Atlassian, 2026.